

Combinatorics PhD Exam, January 2022

1. Prove that if every vertex of a graph G has degree at least 3, then G has a cycle with a chord. (A chord is an edge joining two non-adjacent vertices on the cycle.)
2. Let T be the set of (not rooted) trees with labeled leaves for which each vertex has degree 1 or 3, and let $T_n \subseteq T$ be the subset with exactly n leaves. Show that $T \in T_n$ has exactly $2n - 3$ edges and that $|T_n| = (2n - 5)!! = 1 \cdot 3 \cdot 5 \cdots (2n - 5)$.
3. Let Π_n be the set of permutations of $[n]$ such that $\pi(i + 1) \leq \pi(i) + 1$ for $i \in [n - 1]$. Find $|\Pi_n|$.
4. Recall that König's theorem states that in a bipartite graph G , the maximum size of a matching is equal to the minimum size of a vertex cover. Prove König's theorem from Hall's marriage theorem.
5. Let $(L, \leq)$ be a finite distributive lattice and $\text{Irr}(L)$ be the set of join-irreducibles in L . For $t \in L$, define $K_t = \{p \in \text{Irr}(L) : p \leq t\}$. Show

$$t = \bigvee_{p \in K_t} p.$$

6. Let B be the set of words in the alphabet $\{a, b, c\}$ so that the number of a 's is even and the number of c 's is odd. If b_n is the number of words in B with n letters, find

$$B(x) = \sum_{n \geq 0} b_n x^n.$$

7. Let Λ be the set of integer partitions. Prove the Cauchy identity

$$\prod_{i,j \geq 1} (1 - x_i y_j)^{-1} = \sum_{\lambda \in \Lambda} m_\lambda(x) h_\lambda(y)$$

where $m_\lambda(x)$ is the monomial symmetric function in the variables $x_1, x_2, \dots$ and $h_\lambda(y)$ is the complete homogeneous symmetric function in the variables $y_1, y_2, \dots$.

8. Prove that every permutation of size $n > r \cdot s$ contains either an increasing subsequence of length r or a decreasing subsequence of length s .